GW170817 confines the vestigial-second-sound graviton identification to one part in 10^{15} of its natural range

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The conservative GW170817 cap $|\Delta c/c| \le \tau = 3 \times 10^{-15}$ confines $\chi_{\rm vest} \ge 0$ to $[(1-\tau)^2, (1+\tau)^2]$ under the assumed leading-order law $c_{\rm GW} = c\sqrt{\chi_{\rm vest}}$ and source-emission assumptions. Here c is the electromagnetic reference speed; the free susceptibility $\chi_{\rm vest}$ models normalized dimensionless metric response. The interval has width 4τ, contains 1, and is over fourteen orders of magnitude narrower than adopted $[0.1, 10]$. Microscopic mode, normalization and range are not derived; only this kinematic model is constrained.

INTRODUCTION

Spinor-gravity constructions use composite vielbeins built from fermions [6, 8]. Volovik proposes a vestigial phase with metric order preceding a nonzero tetrad expectation [1]. His symmetry-breaking analysis distinguishes absorbed Nambu–Goldstone modes from the Higgs-sector gravitational-wave modes, whose counting also uses coordinate constraints [2]. Here we assume that a metric-channel excitation of such a substrate can be identified with a spin-2 second-sound graviton.

Motivation and assumed speed law. Volovik applies Landau two-fluid hydrodynamics to a de Sitter state in [3]. The underlying thermodynamic framework treats the gravitational normal component as stiff matter [4]. Within that de Sitter model, the Landau formula and the stiff-component thermodynamic relations give a second-sound speed $s_2 = c$, independently of the de Sitter Hubble parameter H. He discusses several possible graviton interpretations of this collective mode and leaves its type open. For the spin-2 substrate identification assumed here, we also assume the susceptibility-dependent speed law:

$$c^2_{\rm GW}(\chi_{\rm vest}) = c^2 \cdot \chi_{\rm vest}, \qquad c_{\rm GW}(\chi_{\rm vest}) = c\sqrt{\chi_{\rm vest}}, \quad (1)$$

where $\chi_{\rm vest}$ is the dimensionless, normalized metric-channel response of the assumed vestigial substrate. It is a free parameter in this calculation. The random phase approximation (RPA) form in `VestigialSusceptibility.lean` models it using unevaluated bare susceptibility and coupling inputs, so that form alone supplies no numerical value of $\chi_{\rm vest}$. The underlying lattice-fermion framework is the one whose unitarity Vergeles establishes [10]. Equation (1) is an assumed kinematic speed law. We constrain its free susceptibility without deriving a propagating second-sound mode from a microscopic action, its propagator, or its coupling to matter. This missing dynamical bridge is a limitation of the present calculation, not an impossibility result.

The 2017 binary-neutron-star merger GW170817 [11] yielded a tight propagation constraint under the collaboration's conservative distance and emission-time assumptions. Comparing the gravitational-wave arrival time at LIGO/Virgo with the gamma-ray signal at Fermi/INTEGRAL gives $-3 \times 10^{-15} \le \Delta c/c \le +7 \times 10^{-16}$ under those assumptions. We use its conservative symmetric envelope, $|\Delta c/c| \le \tau$ with $\tau = 3\times10^{-15}$. This cap inherits the assumed source-emission window; alternative emission histories can weaken the inferred propagation constraint.

Our assumed speed law satisfies the GW170817 bound only on a sub-interval of our adopted range $\chi_{\rm vest} \in [0.1, 10]$ narrower than that range by more than 14 orders of magnitude. The GW170817-compatible window $[(1-\tau)^2, (1+\tau)^2]$ has width 4τ in $\chi_{\rm vest}$ and sits strictly interior to our adopted range $[0.1, 10]$ about the anchor $\chi_{\rm vest} = 1$. Its relative width is $4\tau/9.9 \approx 1.2 \times 10^{-15}$; a probability mass additionally requires a specified prior. The result is a fine-tuning constraint on one specific kinematic identification, not a refutation of emergent gravity nor of the Akama–Diakonov–Wetterich (ADW) framework. Three possible mechanisms are discussed below.

Relation to the 2017 constraint sweep. Using GW170817 to eliminate a gravity model is not new. Within months of the detection, four Letters analyzed constraints on specified gravitational actions and cosmological models. Baker *et al.* [12] constrained scalar-tensor and vector-tensor theories and obtained a conditional graviton-mass bound in bimetric gravity. Creminelli and Vernizzi [13] required luminal propagation to remain robust on nearby cosmological backgrounds to avoid tuning, reducing the three independent quartic and quintic beyond-Horndeski functions to one. Sakstein and Jain [14] combined the GW bound with cluster and stellar tests to constrain beyond-Horndeski parameters; their separate combination of GW and M87 black-hole constraints made the covariant quartic galileon cosmologically irrelevant for the order-unity matter couplings considered. Ezquiaga and Zumalacárregui [15] also discussed constraints on GW-speed-modifying couplings in Einstein-aether, some sectors of Hořava gravity, and Generalized Proca theories, and the impact on TeVeS and MOND-like theories.

These analyses constrain specified actions, EFT operators and cosmological models under assumptions about backgrounds and matter coupling. This Letter instead constrains the susceptibility in the assumed $c_{\rm GW} = c\sqrt{\chi_{\rm vest}}$ identification. It supplies no microscopic Lagrangian for that identification and constrains no microscopic operator coefficients. Its roughly one-part-in-10^{15} width comparison within the adopted order-unity window is neither a prediction of $\chi_{\rm vest}$ nor a prior-independent probability.

A microscopic realization must determine whether its operator-level mechanisms preserve or alter the assumed speed law and its applicability. The possibilities in the Discussion could fix $\chi_{\rm vest}$ within the compatible interval or modify the identification or its leading-order assumptions. A fixed-point mechanism could preserve the law; the finite-tolerance constraint alone derives none of these mechanisms.

FORMAL SETUP AND RESULT

Vestigial-susceptibility natural range. Our substrate models the vestigial-phase metric-channel susceptibility by the RPA *resummation* $\chi_{\rm RPA} = \chi_0/(1-\gamma_\star\chi_0)$ in the standard Hertz–Millis convention. Here χ_0 is the bubble integral; $\chi_{\rm RPA}$ is the resummed response and $\gamma_\star$ the effective coupling; physical normalization remains undetermined. `VestigialSusceptibility.lean` formalizes that closed form and its stability properties; it evaluates neither χ_0 nor $\gamma_\star$, so it fixes no numerical $\chi_{\rm vest}$, which is why $\chi_{\rm vest}$ enters Eq. (1) as a free parameter. The natural range $\chi_{\rm vest} \in [0.1, 10]$ is an *adopted* naturalness window: the one-loop bubble normalization is $\mathcal{O}(\Lambda_{\rm UV}^2/16\pi^2)$ by dimensional analysis, where $\Lambda_{\rm UV}$ is the ultraviolet cutoff, and we take the $\pm$1-decade window about the resulting unit-normalized central value. We emphasize the provenance: the window is a modeling choice, not a published derivation: the Vergeles lattice analysis [10] establishes unitarity of the underlying Einstein–Cartan–Palatini lattice theory but does not itself compute $\chi_{\rm vest}$ or assign it a range. The comparison window also occurs in the companion study of the linearized Einstein equations [16] for $\alpha_{\rm ADW}$, its free dimensionless coefficient multiplying an adopted induced-Newton baseline. We use that study's $\alpha_{\rm ADW} \in [0.1, 10]$ range as an order-of-magnitude comparison, without identifying $\alpha_{\rm ADW}$ with $\chi_{\rm vest}$ or deriving either range.

The leading-order $c_{\rm GW}$. Under the assumed law (1), a positive susceptibility and physical reference speed $c > 0$ give $c_{\rm GW} > 0$. For every $c \neq 0$, the relative speed deviation is

$$\Delta c/c(\chi_{\rm vest}) \equiv \frac{c_{\rm GW}(\chi_{\rm vest}) - c}{c} = \sqrt{\chi_{\rm vest}} - 1.$$

On the nonnegative domain this deviation is strictly increasing and vanishes if and only if $\chi_{\rm vest} = 1$. Thus unity is the exactly luminal anchor; finite-tolerance compatibility requires an interval rather than this single value.

Finite-tolerance compatibility. Fix $\tau = 3 \times 10^{-15}$ and write $\texttt{LigoSatisfied}(\delta) \equiv |\delta| \leq \tau$ for the conservative symmetric cap.

Theorem 1 (`c_GW_match_iff_chi_close_to_one`)**.** *For $\chi_{\rm vest} \geq 0$, $\texttt{LigoSatisfied}(\Delta c/c(\chi_{\rm vest})) \iff \chi_{\rm vest} \in [(1-\tau)^2, (1+\tau)^2]$.*

Indeed, the cap is equivalent to $1-\tau \leq \sqrt{\chi_{\rm vest}} \leq 1+\tau$. Since $0 < \tau < 1$, both bounds are nonnegative, so squaring gives the stated interval. Conversely, monotonicity of the square root recovers the cap from these squared bounds.

Every compatible susceptibility therefore satisfies

$$0.1 < (1-\tau)^2 \leq \chi_{\rm vest} \leq (1+\tau)^2 < 10.$$

The compatible interval contains $\chi_{\rm vest} = 1$ and has nonzero width exactly $(1+\tau)^2 - (1-\tau)^2 = 4\tau = 1.2 \times 10^{-14}$. The adopted range has width 9.9, and the strict comparison is

$$10^{14}\left[(1+\tau)^2 - (1-\tau)^2\right] < 9.9.$$

This is a comparison of interval widths within an adopted domain. It supplies no probability without a prior: under a uniform prior on $[0.1, 10]$, the compatible interval has probability mass $4\tau/9.9$.

Endpoint checks. The endpoints provide quantitative checks of the confinement.

Theorem 2 (`natural_lower_violates_ligo`)**.** *At $\chi_{\rm vest} = 0.1$, $\Delta c/c = \sqrt{0.1} - 1 \approx -0.6838$, hence $|\Delta c/c|/\tau \approx 2.28 \times 10^{14}$ and* `LigoSatisfied` *is false.*

Theorem 3 (`natural_upper_violates_ligo`)**.** *At $\chi_{\rm vest} = 10$, $\Delta c/c = \sqrt{10} - 1 \approx +2.1623$, hence $|\Delta c/c|/\tau \approx 7.21 \times 10^{14}$ and* `LigoSatisfied` *is false.*

The exclusions already follow from $\sqrt{1/10} \leq \sqrt{1/4} = 1/2$ and $\sqrt{10} \geq \sqrt{4} = 2$. Both endpoints therefore fail simultaneously, while the finite interval in Theorem 1 remains compatible. Figure 1 shows this distinction.

Endpoint constraints and the open dynamical bridge. The endpoint exclusion also applies directly to the relative deviation of the speed itself:

$$\left|\frac{c_{\rm GW}(\chi_{\rm vest}) - c}{c}\right| > \tau, \qquad \chi_{\rm vest} \in \{0.1, 10\}, \quad c \neq 0.$$

This is the statement of `natural_endpoints_violate_ligo_for_every_c`; it includes the physical domain $c > 0$ and follows from the relative-deviation identity above. Changing c therefore cannot make either endpoint compatible under the assumed law.

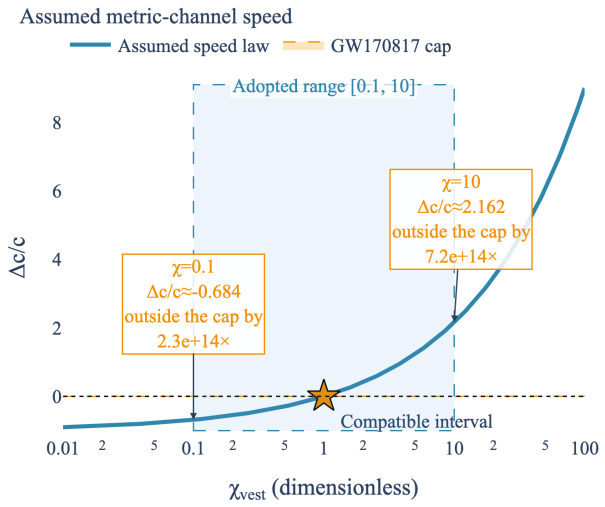


FIG. 1. **Headline result.** The $\Delta c/c = \sqrt{\chi_{\rm vest}} - 1$ deviation across the natural $\chi_{\rm vest} \in [0.1, 10]$ range (pale blue band, dashed blue border) versus the conservative symmetric envelope of the GW170817 [11] multi-messenger constraint, $|\Delta c/c| \leq 3 \times 10^{-15}$ (amber, drawn at $\Delta c/c = 0$ because the cap's own width is far below one pixel at this scale). Over the natural range $\Delta c/c$ sweeps $[-0.68, +2.16]$, exceeding the cap by $\sim 10^{14}\times$ *at the endpoints* (orange callouts) and vanishing at $\chi_{\rm vest} = 1$. The compatible $\chi_{\rm vest}$ window $[(1-\tau)^2, (1+\tau)^2]$ has width $4\tau = 1.2 \times 10^{-14}$, compared with the adopted range's width 9.9. Its finite extent is unresolved at this scale; the gold star marks its anchor $\chi_{\rm vest} = 1$, not a singleton compatible set. The window sits *inside* the band: the datum confines $\chi_{\rm vest}$, it does not exclude the range.

Physical scope. This two-point exclusion leaves the compatible interior interval intact. Neither it nor the c-free endpoint statement quantifies over microscopic actions, propagators or matter couplings, so neither establishes the absence or impossibility of a dynamical derivation. A physical application still requires the response normalization and domain, the physical state, a propagating tensor mode relative to the same matter/light reference, and controlled transport and frequency matching with units and error bounds.

DISCUSSION

What this constrains. The conservative symmetric cap restricts the assumed leading-order identification to $\chi_{\rm vest} \in [(1-\tau)^2, (1+\tau)^2]$. Expanding the endpoints gives $1 - 2\tau + \tau^2$ and $1 + 2\tau + \tau^2$: the susceptibility offsets are approximately $\pm 6 \times 10^{-15}$, whereas the speed-deviation tolerance is $\pm 3 \times 10^{-15}$. Exact luminality occurs at $\chi_{\rm vest} = 1$, but the finite measurement tolerance permits a nonzero-width interval. Its small relative width within our adopted range does not alone establish a prior probability, rule out generic UV theories, or require an exact fixed point at unity.

Possible mechanisms. The finite interval leaves three possibilities open:

1. **Symmetry-driven $\chi_{\rm vest} \to 1$ fixed point.** If a deep-infrared symmetry (e.g. Lorentz invariance of the emergent metric sector) makes $\chi_{\rm vest} = 1$ an exact fixed point of the RG flow, the fine-tuning is structural rather than tuned. An exact fixed point at $\chi_{\rm vest} = 1$ would lie inside the compatible interval. The finite-tolerance constraint alone does not select this mechanism or establish that a microscopic model realizes it. Whether such a fixed point occurs in the ADW broken phase remains a microscopic question.

2. **Distinct metric-channel identification.** We propose as an alternative that the "vestigial mode" is a separate ultraviolet-input metric channel rather than the second-sound degree of freedom. Its dynamics could in principle pin its susceptibility to unity; this remains a model-building possibility. The missing dynamical bridge in this Letter does not establish that this alternative is more likely. Distinguishing the alternatives remains a microscopic question, to be investigated through the nonlinear emergent action.

3. **Frequency-dependent $\chi_{\rm vest}$.** Equation (1) is an assumed frequency-independent kinematic law. A frequency-dependent susceptibility or a compensating correction in the LIGO band is a possible phenomenological extension. We consider the algebraic ansatz $\delta_{\rm disp} = \gamma\omega$, without asserting that it is the leading term of a matched microscopic derivative expansion. The formal probe is the real number $\omega = 100$. Interpreting it as a 100 Hz band requires a physical frequency convention and unit normalization. Identifying $\gamma = \Gamma_H / c_{\rm GW}^2$ introduces a proposed, unmatched dissipative parameter Γ_H in this metric channel. The horizon damping rate denoted by the same symbol in the acoustic setting does not establish that physical identification. Under the assumed coefficient bound $|\gamma| \leq 10^{-30}$, $|\delta_{\rm disp}|$ at this numerical probe is below the GW170817 tolerance. The adopted 10^{-30} is a placeholder chosen to keep the correction below the cap; it is neither derived nor measured. In the further additive model $\delta_{\rm tot} = \delta_0 + \delta_{\rm disp}$, with $\delta_0 = \sqrt{\chi_{\rm vest}} - 1$, that assumed coefficient bound gives $|\delta_{\rm disp}| \leq B = 10^{-28}$ at $\omega = 100$. The reverse triangle inequality then excludes restoration of $|\delta_{\rm tot}| \leq \tau$ whenever $|\delta_0| > \tau + B$. Offsets just outside $\pm\tau$ can nevertheless become compatible through an oppositely signed correction within this bound. Larger corrections are not excluded by the placeholder theorem. Deriving Γ_H and the normalization of this ansatz from

microscopic dynamics remains open.

Comparison with the induced Newton coupling. The in-preparation companion study [16] uses the emergent Newton-coupling parameterization $G_{\mathrm{N,emerg}} = \alpha_{\mathrm{ADW}}\, 12\pi/(N_f \Lambda_{\mathrm{UV}}^2)$, where N_f is a fermion-species-count parameter. In units $\hbar = c = 1$, the observed unreduced Planck mass is $M_{\mathrm{Pl}} = (G_{\mathrm{N,obs}})^{-1/2}$, with $G_{\mathrm{N,obs}}$ the observed Newton constant. At the adopted cutoff $\Lambda_{\mathrm{UV}} = M_{\mathrm{Pl}}$, the exact algebraic match condition is $\alpha^{\star}_{\mathrm{ADW}} = N_f/(12\pi)$. For the companion's stated count choices, the reported rounded matching values span $[0.40, 1.27]$, inside its adopted $[0.1, 10]$ range. Its kernel's Dirac-count convention and these Weyl-count choices require reconciliation before this comparison supplies a physical Standard Model normalization.

The companion proves the matching locus and properties of a candidate coefficient function; it does not extract the microscopic coefficient that selects the locus. That coefficient and χ_{vest} describe distinct response normalizations. Our adopted interval contains unity but derives no value of χ_{vest}. The GW170817 cap confines this parameter to the much narrower compatible interval without supplying a microscopic selection mechanism.

Formal checks. The proposed identification imposes positivity (P1), compatibility with the cap (P2), and sub-half-deviation (P3'):

$$\begin{aligned}&\texttt{H_VestigialModeIsGraviton}(\chi_{\mathrm{vest}})\\ &\quad\equiv \chi_{\mathrm{vest}} > 0 \ \wedge\ \texttt{LigoSatisfied}(\Delta c/c(\chi_{\mathrm{vest}}))\\ &\qquad\wedge\ |\Delta c/c(\chi_{\mathrm{vest}})| < 1/2.\end{aligned}$$

These three clauses are not independent. At the fixed tolerance, P2 implies P3' because $\tau < 1/2$. On the nonnegative domain, Theorem 1 also places every P2-compatible susceptibility in a strictly positive interval, so P1 holds there. Consequently, the complete compatible set for these conditions is precisely that same finite interval, including the anchor $\chi_{\mathrm{vest}} = 1$.

Four counterexamples distinguish compatibility from positivity alone. The endpoints 0.1 and 10 refute the conjunction through failure of P2. At zero, P1 fails; this point lies outside the adopted range. At the interior point $\chi_{\mathrm{vest}} = 1/4$, $\Delta c/c = -1/2$: P1 holds, but P2 and the strict P3' inequality both fail. The quarter-point example therefore shows that P1 alone does not imply P3'. These counterexamples do not establish independence of the three clauses or isolate a uniquely failing clause. The corresponding definitions and algebraic results are recorded in `GravitationalWaves.lean`, Sections 1–4.

For a positive numerical probe ω, `dispersion_within_ligo_iff` gives $|\gamma\omega| \le \tau \iff |\gamma| \le \tau/\omega$ at the fixed GW170817 tolerance τ, bounding the correction alone. `dispersion_correction_abs_bound`, `dispersion_correction_linear_in_gamma` and `dispersion_correction_zero_at_no_dissipation` describe the product's magnitude, additivity in γ and zero-coefficient limit.

`vestigial_dispersion_below_ligo_at_inspiral_peak` records the conditional bound at the numerical probe used above.

Reproducibility. The result is reproducible in two layers. The Lean layer is `GravitationalWaves.lean` (32 theorems, zero `sorry`, zero new axioms, zero `maxHeartbeats` overrides) at the project's pinned Mathlib4 commit; the Python layer is `src/gravitational_waves/` with 89 `pytest` cases cross-checking every numeric quoted in the abstract and figure caption. The headline figure (`src/core/visualizations.py:fig_c_GW_vs_ligo_constraint`) and its underlying numerics are committed-pinned.

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